

# Dedicated built-in-SM $d\bar{d} \rightarrow Z + \text{gluon}$ performance

Apple M3 Pro investigation

pyAmpliCol performance study F

Measured source 202b3fd91a40a6e0422320053796fd71b5b2cd48

This controlled snapshot publishes the authenticated investigation-F table state at wrap-up. Rows  $n = 1\text{--}7$  retain their immutable pre-study evidence and original resource ABI; legacy RSS-only observations are not relabelled as Darwin physical-footprint or exact-decimal-30GB evidence. No new F-contract  $n = 8$  or  $n = 9$  measurement was launched in this run, so unrun cells remain N/A. Catalog-declared ASM O3 and C++ O3 rows above  $n = 6$  remain static N/A without an attempt or compiler launch.

Any future new or repeated  $n = 8\text{--}9$  attempt accepted under study F must be guarded at exactly 30,000,000,000 bytes by the authenticated maximum of aggregate process-tree RSS and Darwin physical footprint, with its authenticated generation phase capped at 3600 seconds. Selected-flow rows use topology-replay with a helicity sum; all-flow rows use all-flow-union with one deterministic helicity. Runtime values are validated at the canonical deterministic phase-space point before timing. Every displayed pyAmpliCol mode reports its authenticated evaluator-total timing where that evidence exists.

## 6.1 Built-in SM dedicated $d\bar{d} \rightarrow Z + \text{gluon}$ performance

### Built-in SM $d\bar{d} \rightarrow Z + \text{gluon}$ ladder (block 1 of 3)

| n setup              | selected flow, helicity sum |               |                    | all flows, single helicity |                |                    |
|----------------------|-----------------------------|---------------|--------------------|----------------------------|----------------|--------------------|
|                      | gen [s]                     | wall [us/pt]  | eval total [us/pt] | gen [s]                    | wall [us/pt]   | eval total [us/pt] |
| 1 AmpliCol reference | 2.18                        | 0.0217        | NOT EXPOSED        | 5.3e-05                    | 0.0475         | NOT EXPOSED        |
| 1 compiled JIT O1    | 2 (x0.919)                  | 0.181 (x8.33) | T 0.181            | C 2; N.C.                  | 0.086 (x1.81)  | T 0.086            |
| 1 compiled ASM O3    | 4.71 (x2.16)                | 0.178 (x8.21) | T 0.178            | C 3.83; N.C.               | 0.0852 (x1.8)  | T 0.0852           |
| 1 compiled C++ O3    | 4.59 (x2.11)                | 0.181 (x8.35) | T 0.181            | C 3.66; N.C.               | 0.083 (x1.75)  | T 0.083            |
| 1 compiled JIT O3    | 1.99 (x0.911)               | 0.174 (x8.04) | T 0.174            | C 1.96; N.C.               | 0.0847 (x1.78) | T 0.0847           |
| 1 eager-DAG JIT O2   | 2.26 (x1.04)                | 0.288 (x13.3) | T 0.288            | C 2.2; N.C.                | 0.181 (x3.81)  | T 0.181            |
| 1 recurrence JIT O2  | 3.97 (x1.82)                | 0.168 (x7.74) | T 0.168            | C 3.81; N.C.               | 0.108 (x2.28)  | T 0.108            |
| 2 AmpliCol reference | 2.18                        | 0.126         | NOT EXPOSED        | 0.000104                   | 0.115          | NOT EXPOSED        |
| 2 compiled JIT O1    | 2.2 (x1.01)                 | 0.39 (x3.09)  | T 0.39             | C 1.91; N.C.               | 0.145 (x1.26)  | T 0.145            |
| 2 compiled ASM O3    | 6.44 (x2.95)                | 0.418 (x3.32) | T 0.418            | C 4.53; N.C.               | 0.129 (x1.12)  | T 0.129            |
| 2 compiled C++ O3    | 6.77 (x3.1)                 | 0.427 (x3.39) | T 0.427            | C 4.46; N.C.               | 0.13 (x1.13)   | T 0.13             |
| 2 compiled JIT O3    | 2.26 (x1.04)                | 0.363 (x2.88) | T 0.363            | C 2.01; N.C.               | 0.137 (x1.19)  | T 0.137            |
| 2 eager-DAG JIT O2   | 2.26 (x1.04)                | 0.685 (x5.43) | T 0.685            | C 2.16; N.C.               | 0.294 (x2.55)  | T 0.294            |
| 2 recurrence JIT O2  | 3.77 (x1.73)                | 0.446 (x3.54) | T 0.446            | C 3.71; N.C.               | 0.283 (x2.46)  | T 0.283            |
| 3 AmpliCol reference | 2.08                        | 0.447         | NOT EXPOSED        | 0.000294                   | 0.312          | NOT EXPOSED        |
| 3 compiled JIT O1    | 2.46 (x1.18)                | 0.703 (x1.57) | T 0.703            | C 2.11; N.C.               | 0.281 (x0.901) | T 0.281            |
| 3 compiled ASM O3    | 13.8 (x6.65)                | 1.1 (x2.47)   | T 1.1              | C 8.11; N.C.               | 0.258 (x0.828) | T 0.258            |
| 3 compiled C++ O3    | 13.7 (x6.62)                | 1.08 (x2.41)  | T 1.08             | C 8.1; N.C.                | 0.26 (x0.835)  | T 0.26             |
| 3 compiled JIT O3    | 2.46 (x1.19)                | 0.646 (x1.44) | T 0.646            | C 2.11; N.C.               | 0.258 (x0.829) | T 0.258            |
| 3 eager-DAG JIT O2   | 2.15 (x1.04)                | 1.41 (x3.14)  | T 1.41             | C 2.8; N.C.                | 0.629 (x2.02)  | T 0.629            |
| 3 recurrence JIT O2  | 3.75 (x1.81)                | 1.18 (x2.63)  | T 1.18             | C 3.73; N.C.               | 0.887 (x2.85)  | T 0.887            |

Original AmpliCol is the denominator. Each pyAmpliCol row reports separate topology-replay and all-flow-union generation and runtime measurements. Parenthesized values are candidate/reference ratios. All-flow generation shows the absolute pyAmpliCol value marked n.c.; n.c. means not comparable because its setup boundary differs from the reference. The wall time is the common runtime observable. Not exposed means that a successful wall measurement has no separately reported evaluator-total timing. Every pyAmpliCol row shows the authenticated accumulated warmed evaluator total marked T; T is not an attribution ratio. Recurrence core/execution attribution remains a separate metric in raw evidence and is not relabeled as evaluator total. Older entries without authenticated total evidence remain marked not exposed; it is not a missing measurement. user cap: native C++/ASM generation is not attempted above n=6 (native-backend-generation-cap-n6-v1).

Built-in SM  $d\bar{d} \rightarrow Z + \text{gluon ladder}$  (block 2 of 3)

| n setup              | selected flow, helicity sum |              |                    | all flows, single helicity |                |                    |
|----------------------|-----------------------------|--------------|--------------------|----------------------------|----------------|--------------------|
|                      | gen [s]                     | wall [us/pt] | eval total [us/pt] | gen [s]                    | wall [us/pt]   | eval total [us/pt] |
| 4 AmpliCol reference | 2.16                        | 1.41         | NOT EXPOSED        | 0.00148                    | 1.15           | NOT EXPOSED        |
| 4 compiled JIT O1    | 3.8 (x1.76)                 | 1.91 (x1.36) | T 1.91             | C 2.79; N.C.               | 0.739 (x0.645) | T 0.739            |
| 4 compiled ASM O3    | 36 (x16.7)                  | 4.39 (x3.11) | T 4.39             | C 22.3; N.C.               | 1.15 (x1.01)   | T 1.15             |
| 4 compiled C++ O3    | 35.8 (x16.6)                | 4.43 (x3.14) | T 4.43             | C 21.6; N.C.               | 1.15 (x1)      | T 1.15             |
| 4 compiled JIT O3    | 3.82 (x1.77)                | 1.76 (x1.25) | T 1.76             | C 2.82; N.C.               | 0.692 (x0.604) | T 0.692            |
| 4 eager-DAG JIT O2   | 2.22 (x1.03)                | 4.1 (x2.91)  | T 4.1              | C 2.51; N.C.               | 1.95 (x1.7)    | T 1.95             |
| 4 recurrence JIT O2  | 3.62 (x1.67)                | 2.87 (x2.04) | T 2.87             | C 3.78; N.C.               | 4.35 (x3.8)    | T 4.35             |
| 5 AmpliCol reference | 2.46                        | 4.66         | NOT EXPOSED        | 0.012                      | 6.62           | NOT EXPOSED        |
| 5 compiled JIT O1    | 10.8 (x4.39)                | 7.86 (x1.69) | T 7.86             | C 8.23; N.C.               | 3.62 (x0.546)  | T 3.62             |
| 5 compiled ASM O3    | 141 (x57.1)                 | 17.7 (x3.8)  | T 17.7             | C 111; N.C.                | 14.4 (x2.17)   | T 14.4             |
| 5 compiled C++ O3    | 147 (x59.7)                 | 22.5 (x4.82) | T 22.5             | C 109; N.C.                | 13.4 (x2.02)   | T 13.4             |
| 5 compiled JIT O3    | 11.6 (x4.7)                 | 5.62 (x1.21) | T 5.62             | C 8.1; N.C.                | 3.33 (x0.502)  | T 3.33             |
| 5 eager-DAG JIT O2   | 2.56 (x1.04)                | 12 (x2.57)   | T 12               | C 5.04; N.C.               | 11.2 (x1.69)   | T 11.2             |
| 5 recurrence JIT O2  | 4.07 (x1.65)                | 6.08 (x1.3)  | T 6.08             | C 4.28; N.C.               | 8.3 (x1.25)    | T 8.3              |
| 6 AmpliCol reference | 3.81                        | 14.7         | NOT EXPOSED        | 0.113                      | 42             | NOT EXPOSED        |
| 6 compiled JIT O1    | 66 (x17.3)                  | 16.5 (x1.12) | T 16.5             | C 57.3; N.C.               | 23.6 (x0.561)  | T 23.6             |
| 6 compiled ASM O3    | 883 (x231)                  | 73.8 (x5.02) | T 73.8             | C 647; N.C.                | 174 (x4.13)    | T 174              |
| 6 compiled C++ O3    | 807 (x212)                  | 71.8 (x4.89) | T 71.8             | C 652; N.C.                | 171 (x4.07)    | T 171              |
| 6 compiled JIT O3    | 67.4 (x17.7)                | 15.9 (x1.08) | T 15.9             | C 64.2; N.C.               | 22.1 (x0.527)  | T 22.1             |
| 6 eager-DAG JIT O2   | 2.67 (x0.699)               | 31.3 (x2.13) | T 31.3             | C 55.4; N.C.               | 81 (x1.93)     | T 81               |
| 6 recurrence JIT O2  | 4.16 (x1.09)                | 14.9 (x1.01) | T 14.9             | C 5.43; N.C.               | 43.4 (x1.03)   | T 43.4             |

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Built-in SM  $d\bar{d} \rightarrow Z + \text{gluon ladder}$  (block 3 of 3)

| n setup              | selected flow, helicity sum |              |                    | all flows, single helicity |              |                    |
|----------------------|-----------------------------|--------------|--------------------|----------------------------|--------------|--------------------|
|                      | gen [s]                     | wall [us/pt] | eval total [us/pt] | gen [s]                    | wall [us/pt] | eval total [us/pt] |
| 7 AmpliCol reference | 5.08                        | 37.8         | NOT EXPOSED        | 1.23                       | 315          | NOT EXPOSED        |
| 7 compiled JIT O1    | 17.5 (x3.44)                | 49.4 (x1.31) | T 49.4             | C 646; n.c.                | 258 (x0.819) | T 258              |
| 7 compiled ASM O3    | STATIC N/A                  | STATIC N/A   | STATIC N/A         | STATIC N/A                 | STATIC N/A   | STATIC N/A         |
| 7 compiled C++ O3    | STATIC N/A                  | STATIC N/A   | STATIC N/A         | STATIC N/A                 | STATIC N/A   | STATIC N/A         |
| 7 compiled JIT O3    | 16.6 (x3.27)                | 46 (x1.22)   | T 46               | C 657; n.c.                | 232 (x0.735) | T 232              |
| 7 eager-DAG JIT O2   | 3.35 (x0.659)               | 83.4 (x2.21) | T 83.4             | C 2.96e+03; n.c.           | 905 (x2.87)  | T 905              |
| 7 recurrence JIT O2  | 7.04 (x1.39)                | 37.9 (x1)    | T 37.9             | C 23.8; n.c.               | 319 (x1.01)  | T 319              |
| 8 AmpliCol reference | N/A                         | N/A          | NOT EXPOSED        | N/A                        | N/A          | NOT EXPOSED        |
| 8 compiled JIT O1    | N/A                         | N/A          | N/A                | N/A                        | N/A          | N/A                |
| 8 compiled ASM O3    | STATIC N/A                  | STATIC N/A   | STATIC N/A         | STATIC N/A                 | STATIC N/A   | STATIC N/A         |
| 8 compiled C++ O3    | STATIC N/A                  | STATIC N/A   | STATIC N/A         | STATIC N/A                 | STATIC N/A   | STATIC N/A         |
| 8 compiled JIT O3    | N/A                         | N/A          | N/A                | N/A                        | N/A          | N/A                |
| 8 eager-DAG JIT O2   | N/A                         | N/A          | N/A                | N/A                        | N/A          | N/A                |
| 8 recurrence JIT O2  | N/A                         | N/A          | N/A                | N/A                        | N/A          | N/A                |
| 9 AmpliCol reference | N/A                         | N/A          | NOT EXPOSED        | N/A                        | N/A          | NOT EXPOSED        |
| 9 compiled JIT O1    | N/A                         | N/A          | N/A                | N/A                        | N/A          | N/A                |
| 9 compiled ASM O3    | STATIC N/A                  | STATIC N/A   | STATIC N/A         | STATIC N/A                 | STATIC N/A   | STATIC N/A         |
| 9 compiled C++ O3    | STATIC N/A                  | STATIC N/A   | STATIC N/A         | STATIC N/A                 | STATIC N/A   | STATIC N/A         |
| 9 compiled JIT O3    | N/A                         | N/A          | N/A                | N/A                        | N/A          | N/A                |
| 9 eager-DAG JIT O2   | N/A                         | N/A          | N/A                | N/A                        | N/A          | N/A                |
| 9 recurrence JIT O2  | N/A                         | N/A          | N/A                | N/A                        | N/A          | N/A                |

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